

COURSE MATERIAL

SUBJECT	BASIC CIVIL AND MECHANICAL ENGINEERING (ME23AES101)
UNIT	2
COURSE	B. TECH
DEPARTMENT	CIVIL
SEMESTER	III-I or I-II
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Version	V-1
PREPARED / REVISED DATE	10-06-2025

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1. Course Objectives

The objectives of this course is to

1. Get familiarized with basic Construction Materials; the scope and importance of Civil Engineering specializations.
2. Introduce the preliminary concepts of Structural and Geotechnical Engineering.
3. Acquire preliminary knowledge on Surveying and Transportation Engineering.
4. Get familiarized with the importance of Water Resources and Environmental Engineering.

2. Prerequisites

Students should have knowledge on

1. Basic Mathematics
2. Basic Physics

3. Syllabus

UNIT II

Structural Engineering: Importance- Types of Structures and structural members- Building Components –Building Planning principles.

Geotechnical Engineering: Types of foundations- Functions and Requirement of a good foundation.

4. Course outcomes

1. Understand various sub-divisions of Civil Engineering and to appreciate their role in ensuring better society and the basic characteristics of Construction Materials.
2. Gain knowledge regarding Structural and Geotechnical Engineering.
3. Explain the concepts of surveying and Transportation Engineering, Water Resources and Environmental Engineering.

5. Co-PO / PSO Mapping

Machine Tools	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	P10	PO11	PO12	PSO1	PSO2
CO1	2	2											2	2
CO2	2	2											2	2
CO3	2	2											2	2

6. Lesson Plan

Lecture No.	Weeks	Topics to be covered	References
1	1	Structural Engineering - Importance	T1
2		Types of structures and structural members	T1, R1
3		Building components	T1, R1
4		Building planning principles	T1, R1
5	2	Geotechnical Engineering	T1, R2
6		Types of foundations	T1, R1
7		Deep foundations	T1, R1
8		Functions and requirement of a good foundation	T1, R1
9	3	Discussion of objective type questions & Short answer questions	T1, R1
10		Discussion of Previous year university questions in question papers	T1, R1

7. Activity Based Learning

1. Students can conduct experiments to test the compressive strength, tensile strength, and workability of concrete.
2. Students can investigate the properties of steel, such as its strength, ductility, and corrosion resistance.
3. Students can use surveying equipment, such as tapes or electronic distance measuring instruments, to measure distances.
4. Students can conduct experiments to test the quality of water, including parameters such as pH, turbidity, and dissolved oxygen.
5. Students can investigate the factors that affect traffic flow.

8. Lecture Notes

2.1 STRUCTURAL ENGINEERING - IMPORTANCE

- Structural engineering is a critical discipline within civil engineering, focused on the design and analysis of buildings, bridges, and other structures to ensure they can withstand various forces and loads.
- The primary aim is to ensure that the structure is safe, stable, and able to withstand the various loads and forces that it will encounter during its lifetime.
- A well-designed and properly executed structural system is essential for ensuring the longevity and safety of a building.
- The key reasons why structural engineering is important in civil engineering are the following
 - 1) Load analysis
 - 2) Design and material selection
 - 3) Safety
 - 4) Durability and longevity
 - 5) Building code compliance
 - 6) Cost-effectiveness
 - 7) Disaster Resilience
 - 8) Energy efficiency
 - 9) Collaboration

1) Load analysis

- Structural engineers conduct a thorough analysis of the loads that the building will encounter, including the weight of the building materials, furniture, equipment, and people, as well as the forces of nature such as wind, earthquakes, and snow.
- This information is used to determine the size and strength of the structural elements that must be used in the building's design.

2) Design and material selection

- Based on the load analysis, the structural engineer designs the structural system and selects the appropriate materials.
- This includes choosing the right type of steel, concrete, or timber, and determining the appropriate size and placement of structural elements such as beams, columns, and foundations.

3) Safety

- Structural engineers ensure that structures can withstand natural forces such as earthquakes, wind, and floods, as well as human-made forces like traffic loads and industrial activities.
- This helps prevent catastrophic failures that could lead to loss of life and property.

4) Durability and Longevity

- Proper structural design ensures that buildings and infrastructure remain functional over their intended lifespan with minimal maintenance.
- This includes consideration of material properties, environmental conditions, and potential deterioration over time.

5) Building code compliance

- Structural engineers must ensure that the building design complies with local building codes and regulations.
- These codes specify minimum requirements for the strength and stability of buildings, and the structural engineer must ensure that the building meets or exceeds these requirements.

6) Cost-effectiveness

- The design of the structural system has a significant impact on the overall cost of the building.
- Structural engineers must balance the need for safety and longevity with the cost of materials and construction, and find cost-effective solutions that meet the requirements of the project.

7) Disaster Resilience

By designing structures that can withstand extreme events such as hurricanes, earthquakes, and tsunamis, structural engineers help communities recover more quickly from disasters and reduce the economic impact of such events.

8) Energy efficiency

- Structural engineers also consider the energy efficiency of the building, and design the structure to minimize the transfer of heat and cold.
- This helps to reduce energy costs and improve the overall comfort of the building.

9) Collaboration

- Structural engineers work closely with other members of the design and construction teams, including architects, contractors, and building owners.
- They must be able to communicate effectively, listen to feedback, and collaborate with others to find the best solutions for each project.

2.2 TYPES OF CIVIL ENGINEERING STRUCTURES

There are six types of structures in Civil Engineering.

They are

- 1) Load bearing structure
- 2) Framed structure
- 3) Truss structure
- 4) Shell structure
- 5) Pre-Engineered structure
- 6) Mass structure

1) Load bearing structure

- Load bearing structures in civil engineering are those designed to carry and transfer loads from the superstructure to the foundation and ultimately to the ground.
- For a load bearing structure, the loads from the roof and other floors get transferred to the wall and wall has to transfer these loads as well as self weight to the foundation.
- These structures are crucial for providing stability, support, and safety to buildings and other infrastructure.
- The load-bearing structures work together to distribute loads safely and efficiently, ensuring the stability and integrity of buildings and other constructions.
- The common load bearing structures are
 - (a) Load bearing walls
 - (b) Beams
 - (c) Columns
 - (d) Trusses
 - (e) Slabs
 - (f) Foundation

a) Load bearing walls

- Load-bearing walls are vertical structures that support the weight of the building or other structures above them.
- They transfer these loads to the foundation and distribute them to the ground.
- Load-bearing walls can be made of brick, concrete block, stone, or other materials.
- Most of the exterior walls of a building structure are considered as load bearing.
- Any load bearing wall should have a minimum thickness of 200 mm.

b) Beams

- Beams are the primary load bearing components of a structure.
- Beams are horizontal structural elements that carry loads perpendicular to their longitudinal axis.

- They are commonly used to support the floors, roofs, and ceilings of buildings.
- The capacity of load bearing depends on the depth and width of the beam.
- The beams are subjected to a higher amount of shear and compressive force.
- Beams can be made of wood, steel, concrete, or other materials.

c) Columns

- Columns are the vertical structural elements that support the weight of beams, slabs, and other components above them, transferring the loads to the foundation.
- Columns are one of the important elements in a structure which have an effective role in the transmission of dead and live loads to the foundation.
- The loads carried may be axial or eccentric.
- The design of column is more important than the design of beams and slabs.
- This is because, if one beam fails it will be a local failure of one floor, but if one column fails, it can lead to the collapse of the whole structure.

d) Trusses

- Frameworks composed of triangular elements that support roofs, bridges, or other structures.
- Trusses efficiently transfer loads to the supports, such as columns or walls, while minimizing material usage.

e) Slabs

- A slab is a flat horizontal structural element that provide floor or roof surfaces and transfer loads to beams and columns.
- Slabs usually carry vertical loads.
- Slabs are commonly made of concrete and steel.

f) Foundation

- The lowest part of a building that distributes the weight of the structure evenly onto the soil or bedrock beneath, providing stability and preventing settlement.
- Foundations can be shallow or deep depending on soil conditions and structural requirements.

2) Framed structure

- Framed structures are another type of building structure in civil engineering.
- It's work is based on the structural elements that transfer the loads from slab to footing.
- A framed structure is composed of beams and columns.

- An RCC framed structure is basically an assembly of slabs, beams, columns and foundation interconnected to each other as a unit.
- The load transfer in such a structure takes place from slabs to the beams, from the beams to the columns and then to the lower columns and finally to the foundation.
- The walls are built to partition the living area.
- These walls are subjected to self-weight only.
- This type of structures are required when number of stories in a building is more and also when larger areas are to be covered free from walls.

Types of framed structures

i) Rigid frames

- These frames are built at the site.
- In rigid frame structures, beams and columns are interconnected to form a rigid framework that can resist horizontal loads such as wind and seismic forces.
- Rigid frames are commonly used in buildings, bridges, and industrial structures.
- Rigid frames provide more stability and resist rotations effectively.

ii) Braced Frames

- Braced frames incorporate diagonal or cross-bracing elements to provide lateral stability and resist wind or seismic loads.
- They are commonly used in buildings and structures where lateral loads are a significant design consideration.

3) Truss structure

- A truss is a structure consisting of straight axial members arranged in triangles and connected at pin joints.
- Trusses efficiently transfer loads to the supports, such as columns or walls, while minimizing material usage.
- Trusses are used in a broad range of buildings, mainly where there is a requirement for very long spans, such as in airport terminals, aircraft hangers, sports stadia roofs, auditoriums etc.
- Trusses are also used to carry heavy loads and are sometimes used as transfer structures.

4) Shell structure

- A shell structure, as the name suggests, is characterized by a curved, shell-like form.
- Unlike traditional framed structures, which rely on beams and columns for support, shell structures derive their strength primarily from their shape and geometry.
- They are commonly used in architecture and engineering for their aesthetic appeal and structural efficiency.
- Shell structures require careful design and engineering to ensure structural stability and integrity.

- Factors such as geometry, material selection, loading conditions, and construction techniques must be considered during the design process to achieve optimal performance and durability.

Types of shell structures

- a) Dome
- b) Vault
- c) Shell roofs
- d) Folded plate structures
- e) Shell grid structures

5) Pre-Engineered Structures

- Pre-engineered structures, also known as pre-engineered buildings (PEBs).
- Pre-Engineered Buildings (PEBs) are the building components that are manufactured at a factory and assembled on site.
- PEB structural components are fabricated at the factory to the exact size, transported to the site, and assembled at the site, usually with bolted connections.
- Usually, PEBs are steel structures.
- Pre-engineered structures offer several advantages in terms of cost, time, and flexibility compared to traditional construction methods.

6) Mass structure

- Mass structures in civil engineering refer to large-scale constructions primarily composed of materials like concrete, stone, or earth.
 - These structures are designed with a focus on stability, durability, and functionality.
- Examples : Dams, Retaining walls, tunnels etc.

BUILDING COMPONENTS

The following are the basic components of a building

- 1) Foundation
- 2) Plinth
- 3) Walls and columns
- 4) Sills, lintels and chejjas
- 5) Doors and windows
- 6) Floors
- 7) Roofs
- 8) Steps, stairs and lifts
- 9) Finishing work
- 10) Building services

1. Foundation

- Foundation is the most important part of the building.
- Building activity starts with digging the ground for foundation and then building it.
- It is the lower most part of the building.

- It transfers the load of the building to the ground or soil.
 - It can be shallow or deep depending on soil conditions and building loads.
 - A good foundation prevents settlement of the building.
- Its main functions and requirements are
- (a) Distribute the load from the structure to soil evenly and safely.
 - (b) To anchor the building to the ground so that under lateral loads building will not move
 - (c) It prevents the building from overturning due to lateral forces.
 - (d) It gives level surface for the construction of super structure.

2) Plinth

- The plinth level refers to the base or foundation of a building or structure.
- It is the level at which the ground floor or the base of a building is constructed.
- The primary purpose of the plinth level is to provide a stable and level surface for constructing the walls and floors of a building.
- The plinth is constructed above the ground level.
- It is usually 75 mm thick plain concrete course.
- The function of the plinth is to keep the ground floor above ground level.
- It is required that plinth level is at least 150 mm above the road level, so that connections to underground drainage system can be made.

3) Walls and Columns

- The function of walls and columns is to transfer the load of the structure vertically downwards to transfer it to foundation.
- Walls are masonry structures and Columns are reinforced concrete structures.
- The wall divide the space into different rooms.
- The purposes of walls in buildings are to support roofs, floors and ceilings.
- Columns support structural load and transfer it through beams.

4) Sills, Lintels and Chejjas

a) Sill

- A window sill, also known as a window ledge or window bottom.
- A window sill is the horizontal structure or surface at the bottom of a window.
- Window sills serve to structurally support and hold the window in place.
- The thickness of the window sill is generally 50 mm to 75 mm.
- Window sill is made up of concrete.

b) Lintel

- Window lintels are beams that run along the top of windows in order to support and properly distribute the structure's overall weight.

- The primary function of a lintel is to bear the weight from above, distributing it evenly to the sides of an opening.
- The width of lintels is equal to the width of wall.
- Lintels also provide structural support and prevent deformation of the masonry work or framing around doors and windows.

c) Chejjas

- Chejja is the projection given outside the wall to protect doors and windows from the rain.
- They are usually made with R.C.C.
- In low cost houses stone slabs are provided as chejjas.
- The projection of chejja varies from 600 mm to 800 mm.

5) Doors and Windows

a) Doors

- The function of a door is to give access to different rooms in the building and to deny the access whenever necessary.
- Number of doors should be minimum possible.
- The size of the door should be of such dimension as will facilitate the movement of the largest object likely to use the door.

b) Windows

- Windows are provided to get light and ventilation in the building.
- They are located at a height of 0.75 m to 0.9 m from the floor level.
- In hot and humid regions, the window area should be 15 to 20 per cent of the floor area.

6) Floors

- Floors are the important component of a building.
- Floors are provided to divide a building into different stories for creating accommodation one above the other.
- The main function of the floor is to give support to occupants, furniture's and equipment of a building.
- The floor should be strong enough to carry the loads (self weight and imposed loads).
- The floor should be clean, smooth, impervious, durable and weather resistant.

7) Roof

- Roof is the top most portion of the building which provide top cover to the building.
- It should be leak proof.

- Roof may be flat, sloping or curved type.
- Roof should be strong, stable and weather resistant.
- It should have adequate resistance against fire. It should have sufficient heat and sound insulation.

8) Steps, Stairs and Lifts

a) Steps

- Steps are used for going from one level to another level.
- They are fixed in a position.
- Each step has one tread and one rise.
- The horizontal portion of a step on which a person walks is called tread.
- The vertical portion of a step is called rise.
- Rise and tread are proportioned so as to provide convenient and easy access.
- The rise may vary from 150 mm to 200 mm.
- The tread may vary from 250 mm to 300 mm.
- Number of steps required depends upon the difference in the levels of the ground and the floor.

b) Stairs

- Stairs are a set of steps built between two or more floors in a building.
- Stairs provide access for various floors of the building.
- The stair consists of series of steps with landings at appropriate intervals.
- The stretch between the two landings is called flight.

c) Lifts

- In all public buildings lifts are to be provided for the conveniences of old and disabled persons.
- In hostels G + 3 floors can be built without lifts, but in residential flats maximum floors permitted without lifts is only G + 2.
- Lift is to be located near the entrance.
- Size of the lift is decided by the number of users in peak hours.
- Lifts are available with capacity 4 to 20 persons.

9) Finishing

- Bottom portion of slab (ceiling), walls and top of floor need smooth finishing with plaster.
- Then they are provided with white wash, distemper or paints or tiles.

The function of finishing work is

- (a) Give protective cover
- (b) Improve aesthetic view

10) Building Services

Water supply, sanitation and drainage works, electric supply work and construction of cupboards and show cases constitute major building services.

BUILDING PLANNING

- Building Planning is the arrangement of various components or units of a building in a systematic manner so as to form a meaningful and homogeneous structure to meet its functional purpose.
- The arrangement of the various rooms in the building is known as Planning of Building.
- Building planning is a graphical representation of what a building will look like after construction.
- It is used by builders and contractors to construct buildings of all kinds.
- Building planning is also useful when it is essential to estimate how much a project will cost and for preparing project budgets building planning is also useful.
- The basic objective of planning the building is to arrange all the units of the building on all floors at a given level according to their functional requirements.
- By doing this, one can make the best use of space available for building.

BUILDING PLANNING PRINCIPLES

The following are the principles of building planning

- 1) Aspect
- 2) Prospect
- 3) Furniture Requirements
- 4) Roominess
- 5) Grouping
- 6) Circulation
- 7) Lighting
- 8) Privacy
- 9) Elegance
- 10) Flexibility
- 11) Sanitation
- 12) Economy

1) Aspect

- Aspect is a very important consideration in the planning of a building.
- It influences the appearance of a building.
- Aspect is the positioning of rooms in buildings with respect to 4 directions in such a way that the occupants of buildings would enjoy the natural comforts like sunshine, breeze, scenery, etc. to the maximum possible extent.
- A room receiving light and air from any particular direction is said to have the aspect of that direction.

- Aspect not only provides comfort, but is a requisite from the hygienic point of view as well.

Some necessary aspects of commonly constructed rooms are listed below

A) Living Room

- It should have a southern or southeast aspect.
- The sun is towards the south during winter and north during summer which will provide sunshine during winter and cooler during summertime.

B) Bedroom

It should have a west or south-west aspect, as the breeze required particularly in summer would prevail from this side.

C) Kitchen

It should have an eastern aspect so as to admit morning sun refreshes and purify the air.

D) Gallery or Verandah

It should be north or north-east aspect.

E) Classroom, Reading Room

It should be laid with the north aspect as light received from the north will be diffused and evenly distributed.

2) Prospect

- A building is said to have prospect when it presents a good and pleasing appearance when seen from outside.
- Prospect make outer appearance attractive.
- The outside appearance can be improved by attractive planning, providing bay windows and utilization of good landscape or sea scape.

3) FURNITURE REQUIREMENTS

- Furniture is a functional requirement of a room.
- A living room, drawing room, kitchen, classroom, office room, laboratory, hospital room, etc. all have their own furniture requirements.
- A room should have enough space to accommodate all the furniture required for the maximum number of people without overcrowding.

4) Roominess

- Roominess means getting maximum benefit from the minimum dimension of the room.

- The effect of roominess depend on the relative dimension of length, width and height of the room.
- The shape and size of the room are also very important for roominess.
- A rectangular room has more roominess than square room in same area.

5) GROUPING

- Grouping is concerned with the relation of each room with other rooms of the building.
- Kitchen and dining room should be close to each other.
- The kitchen should be kept away from the main living room.
- Store room should be adjacent to kitchen.
- In an office buildings, hospitals, etc. the administrative department should be located centrally for convenience and economy of services.
- Washrooms should be far away from kitchen.

6) Circulation

- Access or internal connection between rooms on the same floor or between floors is known as circulation.
- Circulation between rooms on the same floor is called horizontal circulation like- passages, corridors, halls, etc.
- Circulation between various floors is known as vertical circulation, like- stairs, lifts, etc.

7) Lighting

- Natural and artificial lighting is provided in budling.
- By proper ventilation sunlight can given good lighting.
- By using various electrical appliances artificial lighting is provided.

8) Privacy

- Privacy is an important part of building planning.
- For residential building planning privacy is important consideration.
- There should be privacy from one room to another room and also from neighbour building.
- Proper grouping of rooms, and good positioning of doors and windows, lobbies, or screens can give required internal privacy.
- Privacy is necessary in the bedroom and washrooms.

9) Elegance

- Elegance refers to the planning of elevation and layout of the plan to give an impressive appearance to the building.
- To get good elegance of a building it is better if elevation is developed first and then the plan is adjusted accordingly.

- The selection of a site for the construction of a building greatly affects elegance.

10) Flexibility

- The planning of the building should be prepared by keeping in mind the future requirement.
- Expansion should be possible economically without major alterations in planning.
- For designing houses for middle-class families or other buildings where the economy is the main consideration flexibility should always be considered.

11) Sanitation

- Sanitation of the building is not only associated with urinals, bathrooms, washbasins, and sinks but also the overall lighting and ventilation.
- Provision for cleanliness, lighting and ventilation in sanitary units avoid growth of bacteria and spread of disease condition.

12) Economy

- The economy is a major factor in building planning.
- Building planning should be carried out in the financial limit of the client.
- An engineer should know in advance what client intends to spend for the building and according material of construction.
- But while considering the economy, the required strength and durability of the structure should not be compromised.

GEOTECHNICAL ENGINEERING**SUBSTRUCTURE**

- A structure essentially consists of two parts, namely the super structure which is above the plinth level and the substructure which is below the plinth level.
- Substructure is otherwise known as the foundation and this forms the base for any structure.
- Generally, about 30 percent of the total construction cost is spent on the foundation.

FOUNDATIONS

- Foundation is an important part of the structure which transfers the load of the super structure to the foundation soil.
- The soil on which the foundation rests is called foundation soil.
- The foundation distributes the load over a larger area, so that pressure on the soil does not exceed its allowable bearing capacity and restricts the settlement of the structure within the permissible limits.
- Foundation increases the stability of the structure.

TYPES OF FOUNDATIONS

Foundations mainly split in to two types.

They are

- 1) Shallow foundations
- 2) Deep foundations

1) SHALLOW FOUNDATIONS

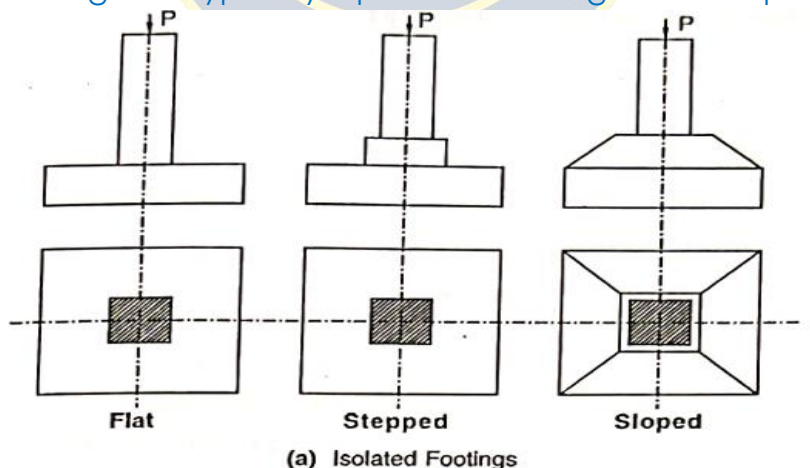
- When the depth of the foundation is less than or equal to its width, it is defined as a shallow foundation.
- Shallow foundations are appropriate when the surface soil is strong and stable enough to handle the weight of the building or structure.
- Shallow foundations are widely used in construction due to their simplicity, cost-effectiveness, and ease of installation.
- They are typically used for residential buildings, small commercial buildings and other light structures.

TYPES OF SHALLOW FOUNDATIONS

- a) Spread footings (Isolated footings)
- b) Combined footings
- c) Strip footing
- d) Mat foundations

a) Spread Footings

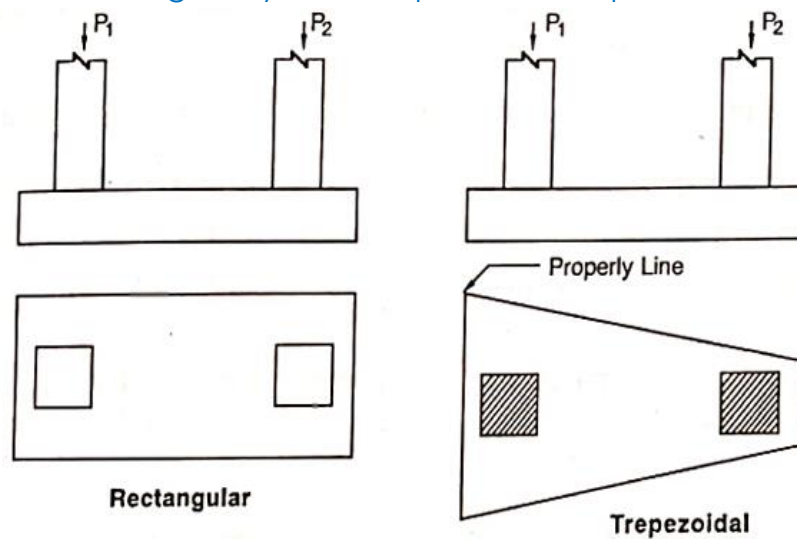
- These are also called isolated footings
- Isolated footings are used for individual columns or loads.
- These footings are typically square or rectangular in shape.



b) Combined Footings

- Combined footings are used for multiple columns or loads.
- These footings combine individual footings into a single unit.
- Combined footing is adopted when the space between two columns is so small.

- Combined footings help distribute the load of multiple columns to the soil below and provides adequate support.
- Combined footings may be in trapezoidal shape or rectangular shape.



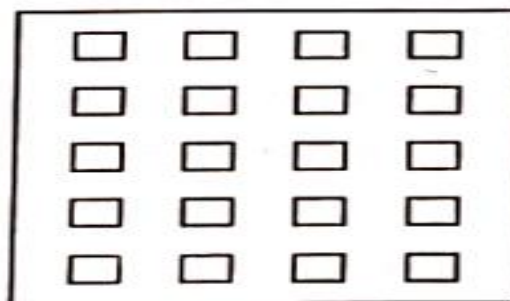
(b) Combined Footings

c) Strip Footings

- Strip footing is one of the different types of shallow foundations.
- Strip footing is used where soil of good bearing capacity is available at a depth of less than 3 m from the ground level.
- Strip footing distribute the building's load evenly across the length of the wall.
- Since the footing is provided throughout the length of the wall in case of load bearing walls, it is also called as wall footing.

d) Mat Foundations

- Mat foundations are also known as raft foundations.
- Mat foundation consists of a large, thick slab of reinforced concrete designed to support the entire load of a building.
- It is typically used when the soil near the surface has low bearing capacity and for large and heavy structures.



(d) Raft Footing

2) Deep Foundations

- A deep foundation is a construction support system that transfers a building's weight to deeper, more stable soil or rock layers.
- Deep foundations are used when the soil does not have sufficient strength or bearing capacity to support the loads of a structure.
- Deep foundations are typically used when the upper soil layers are weak and highly compressible.
- Deep foundations can be more expensive and time-consuming.

TYPES OF DEEP FOUNDATIONS

- a) Pile foundations
- b) Pier foundations
- c) Well foundations
- d) Screw piles

a) Pile Foundations

- A pile foundation is a deep foundation system that uses long, slender columns to support structures on top of weak or unstable soil.
- Piles are long, slender columns made of materials like concrete, steel, or timber that are driven, drilled, or screwed deep into the ground.

Types of Pile Foundations

i) End-Bearing Piles

- These piles transfer the load to a hard layer of soil or rock at the bottom of the pile.
- The load is carried by the tip of the pile (called the "toe") resting on strong, stable material like bedrock or hard soil.
- End-bearing piles are used when solid rock or a firm layer of soil is at a relatively shallow depth.

ii) Friction Piles

- Friction piles rely on the friction between the pile surface and the surrounding soil to carry the load.
- The load is distributed along the length of the pile, and the soil provides resistance to the pile's movement.
- Friction piles are used when there is no strong layer at depth and the load must be transferred through friction with surrounding soil.

iii) Combination Piles

- These piles rely on both the end bearing and friction to support the load.
- They are typically used in intermediate soil conditions where neither end-bearing nor friction alone is sufficient.

b) Pier Foundations

- A pier foundation is a type of deep foundation that consists of vertical columns, or "piers," which are used to support the load of a structure by transferring it to deeper, more stable soil or bedrock.
- The piers are typically made of concrete, steel, or masonry.
- Generally, pier foundation is shallow in depth than the pile foundation.

c) Well Foundations

- Well foundations, also known as well casings or caissons, are a type of deep foundation used in civil engineering to transfer loads from a structure to a deep, stable soil or rock layer.
- Well foundations are constructed by excavating a large, circular or rectangular shaft and then filling it with concrete. The shaft is often lined with a steel casing or reinforced concrete to prevent collapse.
- Well foundations are used when the soil near the surface is unstable or unable to support the weight of the structure. They can extend to depths of 10-50 meters or more, depending on the site conditions.
- Well foundations can support heavy loads, making them suitable for large structures such as bridges, high-rise buildings, and heavy industrial facilities.

d) Screw Piles

- Screw piles, also known as helical piles, are a type of deep foundation system used in construction to support structures.
- They are long, steel shafts with helical (screw-like) blades or plates that are mechanically driven into the ground, similar to a screw being turned into wood.

Functions of Foundations

Foundations are a crucial part of any structure, and their primary functions include

1) Load Transfer

Foundations transfer the weight of the structure to the ground, distributing the load evenly to prevent settlement or collapse.

2) Stability

Foundations provide stability to the structure, preventing it from tilting, settling, or sliding due to external forces like wind, earthquakes, or soil movement.

3) Soil Interaction

Foundations interact with the soil, transferring loads and resisting soil movements, such as settlement or heave.

4) Levelling

Foundations help to level the structure, ensuring it is even and plumb, which is essential for construction and occupancy.

5) Durability

A well-designed foundation can enhance the durability of the structure, protecting it from damage caused by soil settlement, water infiltration, or other environmental factors.

REQUIREMENTS OF GOOD FOUNDATION

- A good foundation is crucial for the stability and longevity of any structure.
- The foundation must safely distribute the weight of the building to the ground, prevent excessive settlement, and resist external forces such as wind, earthquakes, or floods.

The requirements of good foundation are

- 1) Adequate load distribution
- 2) Sufficient bearing capacity
- 3) Resistance to Settlement
- 4) Resistant to External Forces
- 5) Durability and longevity
- 6) Cost-Effectiveness

1) Adequate Load Distribution

- The foundation must be able to distribute the weight of the structure evenly across the ground to prevent localized settlement or tilting.
- It should be designed to handle both the dead load (permanent weight of the building) and live load (temporary loads like people, furniture, and equipment).

2) Sufficient Bearing Capacity

- The foundation must rest on soil or rock that can bear the weight of the structure without undergoing excessive settlement.
- The soil beneath should have a high enough bearing capacity to support the load, which is determined through geotechnical investigation.

3) Resistance to Settlement

- A good foundation must prevent excessive settlement (the gradual sinking or shifting of the foundation into the ground).
- This is important to prevent cracking or tilting of the structure.
- Differential settlement (where different parts of the building settle at different rates) must be minimized.

4) Resistant to External Forces

- The foundation should be designed to resist external forces, such as wind, earthquakes, and floods.
- In seismic zones, foundations must withstand lateral forces to avoid structural damage.
- Adequate drainage provisions (such as proper slope, waterproofing, or drainage pipes) are necessary to prevent water accumulation beneath the foundation.

5) Durability and Longevity

- Materials used for the foundation must be durable and resistant to environmental factors like moisture, soil chemicals, and temperature variations.
- The foundation must be designed to resist corrosion, erosion, or decay, especially in areas with aggressive soils or high-water tables.

6) Cost-Effectiveness

- While it is essential for the foundation to be strong and durable, it must also be cost-effective.
- Over-engineering can result in unnecessary costs, while under-engineering can compromise safety.
- The foundation design should balance safety, performance, and budget.

9. Practice Quiz

1. Civil engineers contribute the following to the welfare of the society

- Planning of towns
- Water supply
- Transportation

d) All of the above

2. _____ engineering deals with the structural design and analysis of buildings, bridges etc.

- Structural**
- Geotechnical
- Transportation
- Environmental

3. _____ engineering deals with the behavior of soils and rocks and design of foundation.

- Structural
- Geotechnical**
- Transportation
- Environmental

4. The branch of engineering that deals with the construction of roads, bridges etc is called _____

- Structural
- Geotechnical
- Transportation**
- Environmental

5. The branch of engineering that deals with the purification of drinking water is called

- Structural
- Geotechnical
- Transportation



d) Environmental**6. Cement was invented in the year**

a) 1624

b) 1724

c) 1824

d) 1924

7. The fineness of the cement can be found out by sieve analysis by using IS sieve number

a) 60 micron

b) 70 micron

c) 80 micron

d) 90 micron**8. According to Indian standards, the size of the brick is (in cm)****a) 19 X 9 X 9**

b) 20 X 9 X 9

c) 19 X 10 X 10

d) 20 X 10 X 10

9. _____ class bricks are used for the construction of load bearing walls.**a) First**

b) Second

c) Third

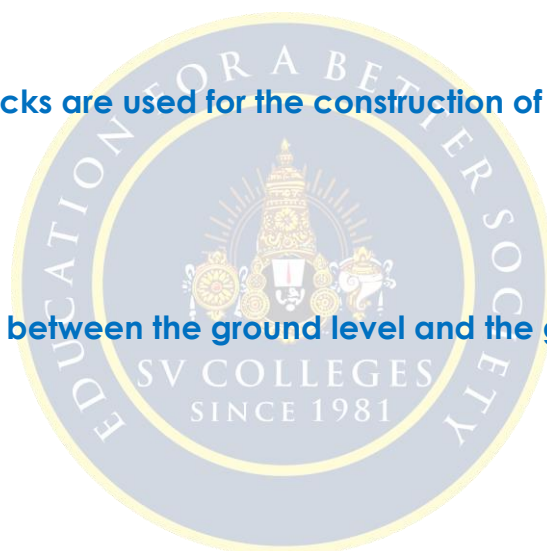
d) Fourth

10. The portion of the wall between the ground level and the ground floor is called**a) Plinth**

b) Lintel

c) Sill

d) Chejjas

**10. Assignments**

S.No	Question	BL	CO
1	Explain the importance of structural engineering.	2	2
2	Explain the different types of civil engineering structures.	2	2
3	Explain the different components of building with neat sketches.	2	2
4	What is meant by building planning? Explain the building planning principles with neat sketches.	2	2
5	What are shallow foundations and explain different types of shallow foundations.	2	2
6	What are deep foundations and explain different types of deep foundations.	2	2

11. Part A- Question & Answers

S.No	Question & Answers	BL	CO
1	List various disciplines of Civil Engineering. Ans. 1) Structural Engineering 2) Geotechnical Engineering 3) Transportation Engineering 4) Hydraulics & Water Resources Engineering 5) Environmental Engineering	1	1
2	What are building materials? Ans. The materials used in the construction of engineering structures such as buildings, bridges and roads are called construction materials or building materials or engineering materials.	1	1
3	What is cement? Ans. Cement is commonly used binding material in construction.	1	1
4	What are the raw materials required for manufacture of Portland cement? Ans. The cement is obtained by burning a mixture of calcareous (calcium) and argillaceous (clay) material at a very high temperature and then grinding the clinker so produced to a fine powder.	1	1
5	What is meant by hydration of cement? Ans. Hydration is the chemical reaction between cement and water by which Portland cement becomes a firm, hardened mass.	1	1
6	List the properties of good sand. Ans. 1) It should be clean and coarse. 2) It should be free from any organic matter or vegetable matter. 3) It should be chemically inert. 4) It should contain sharp, angular, coarse and durable grains.	1	1
7	What are the different uses of bricks? Ans. 1) Bricks are extensively used as a leading material of construction. 2) Bricks are extensively used for construction of load-bearing walls and partition walls.	2	1
8	What is meant by segregation? Ans. Separation of coarse particles from the green concrete is called segregation.	2	1
9	What is meant by bleeding? Ans. This refers to the appearance of the water along with cement particles on the surface of the freshly laid concrete.	2	1
10	What is meant by converted timber? Ans. By sawing, rough timber is converted into various commercial sizes like planks, battens, posts, beams etc. Such form of timber is known as converted timber.	2	1

12. Part B- Questions

S.No	Question	BL	CO
1	Explain the importance of structural engineering.	2	2
2	Explain the different types of civil engineering structures.	1	2
3	Explain the different components of building with neat sketches.	2	2
4	What is meant by building planning? Explain the building planning principles with neat sketches.	2	2
5	What are shallow foundations and explain different types of shallow foundations.	2	2

13. Supportive Online Certification Courses

1. Introduction to Civil Engineering Profession By Prof. Subhadeep Banerjee, conducted by IIT Madras – 8 weeks

14. Prescribed Text Books & Reference Books

Text Book

1. Basic Civil Engineering, M. S. Palanisamy, Tata Mcgraw Hill publications (India) Pvt. Ltd. Fourth Edition.
2. Introduction to Civil Engineering, S.S. Bhavikatti, New Age International Publishers. 2022. First Edition.

References:

1. Hydrology and Water Resources Engineering, Santosh Kumar Garg, Khanna Publishers, Delhi. 2016
2. Highway Engineering, S.K.Khanna, C.E.G. Justo and Veeraraghavan, Nemchand and Brothers Publications 2019. 10th Edition.